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[3] 6s

```python
# Import libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import skew, kurtosis, zscore

# ----------------------------
# 1 Generate Data
# ----------------------------
np.random.seed(42)

normal_data = np.random.normal(loc=50, scale=5, size=1000)
skewed_data = np.random.exponential(scale=5, size=1000)

# ----------------------------
# 2 Visualization
# ----------------------------
plt.figure(figsize=(12,5))

plt.subplot(1,2,1)
sns.histplot(normal_data, kde=True)
plt.title("Normal Distribution")

plt.subplot(1,2,2)
sns.histplot(skewed_data, kde=True)
plt.title("Exponential Distribution (Right-Skewed)")

plt.tight_layout()
```

Variables Terminal ✓ 13:57 Python 3

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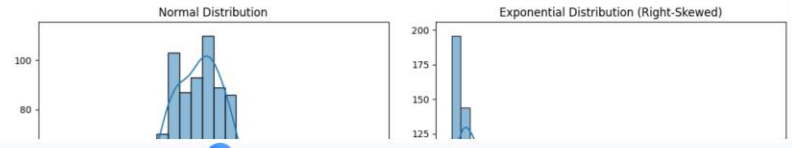
[3] 6s

```python
# ----------------------------
# 3 Calculate Shape
# ----------------------------
print("Normal Data Skewness:", skew(normal_data))
print("Skewed Data Skewness:", skew(skewed_data))

print("\nNormal Data Kurtosis (Fisher's):", kurtosis(normal_data))
print("Skewed Data Kurtosis (Fisher's):", kurtosis(skewed_data))

# ----------------------------
# 4 Outlier Detection (Z-score)
# ----------------------------
def detect_outliers(data):
    z_scores = zscore(data)
    outliers = np.where(np.abs(z_scores) > 3)
    return outliers[0]

outliers = detect_outliers(skewed_data)
print("\nNumber of Outliers in Skewed Data:", len(outliers))
```



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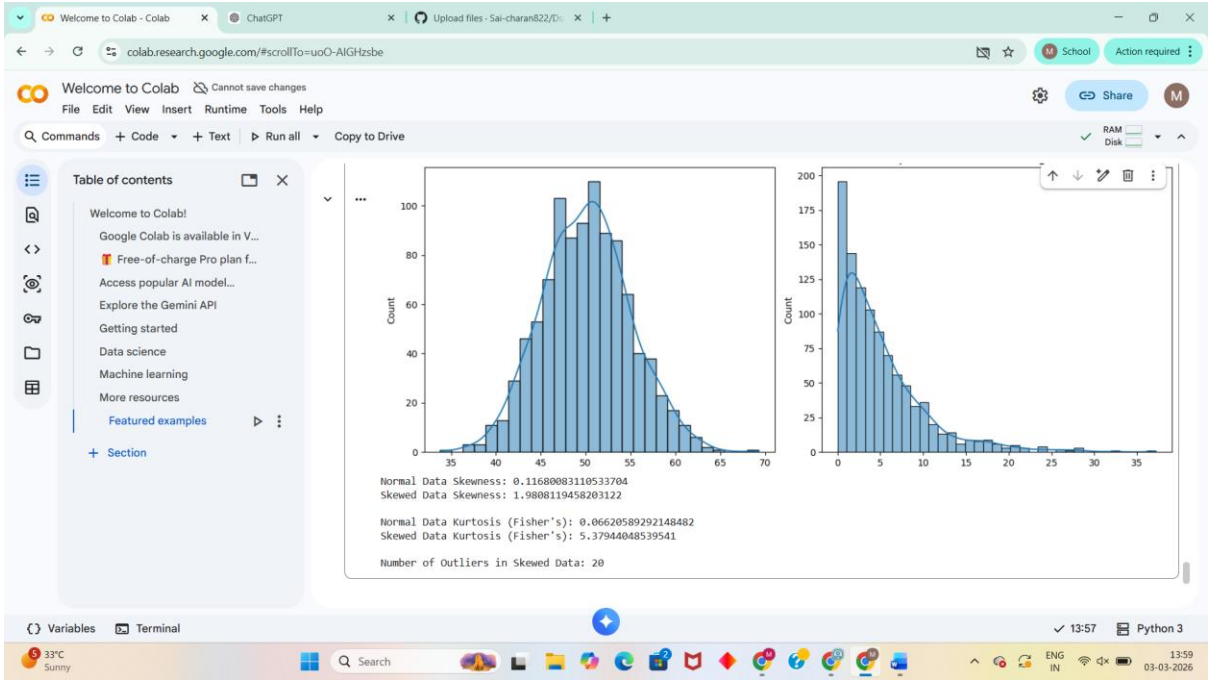
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Count
Normal Data Skewness: 0.11680083110533704
Skewed Data Skewness: 1.9808119458203122
Normal Data Kurtosis (Fisher's): 0.06620589292148482
Skewed Data Kurtosis (Fisher's): 5.37944048539541
Number of Outliers in Skewed Data: 20
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Activity 2

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[5]

```python
# Import libraries
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from scipy.stats import zscore

# ----------------------------
# 1 Create Dataset
# ----------------------------
prices = np.array([2.5, 3.0, 2.8, 3.2, 15.0, 2.9, 3.1, 2.7, 100.0])

df = pd.DataFrame({"House_Price_$100k": prices})

print("Original Data:\n")
print(df)

# ----------------------------
# 2 Identify Outliers (Z-score)
# ----------------------------
df["Z_score"] = zscore(df["House_Price_$100k"])

# Flag |Z| > 3
df["Outlier_Z"] = np.abs(df["Z_score"]) > 3

print("\nData with Z-scores:\n")
print(df)

print("\nNumber of Outliers (Z-score method):"
```

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[5]

```python
print("\nNumber of Outliers (Z-score method):",
      df["Outlier_Z"].sum())

# ----------------------------
# 3 Visualize Outliers (Boxplot)
# ----------------------------
plt.figure(figsize=(6,4))
sns.boxplot(x=df["House_Price_$100k"])
plt.title("Boxplot of House Prices (Before Treatment)")
plt.show()

# ----------------------------
# 4 Treat Outliers
# ----------------------------

# Correct the typo (100.0 → 10.0)
df_corrected = df.copy()
df_corrected.loc[df_corrected["House_Price_$100k"] == 100.0,
                 "House_Price_$100k"] = 10.0

# Optional: Log transformation for handling 15.0 mansion
df_corrected["Log_Price"] = np.log(df_corrected["House_Price_$100k"])

print("\nCorrected Data:\n")
print(df_corrected)

# ----------------------------
# 5 Visualization After Treatment
# ----------------------------
plt.figure(figsize=(6,4))
```

Variables Terminal 14:01 Python 3

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```
plt.figure(figsize=(6,4))
sns.boxplot(x=df_corrected["House_Price_$100k"])
plt.title("Boxplot After Correcting Typo")
plt.show()
```

```
Original Data:

   House_Price_$100k
0               2.5
1               3.0
2               2.8
3               3.2
4              15.0
5               2.9
6               3.1
7               2.7
8             100.0

Data with Z-scores:

   House_Price_$100k   Z_score  Outlier_Z
0               2.5 -0.413529      False
1               3.0 -0.397018      False
2               2.8 -0.403622      False
3               3.2 -0.390413      False
4              15.0 -0.000734      False
5               2.9 -0.400320      False
6               3.1 -0.393715      False
7               2.7 -0.406925      False
8             100.0  2.806275      False
```

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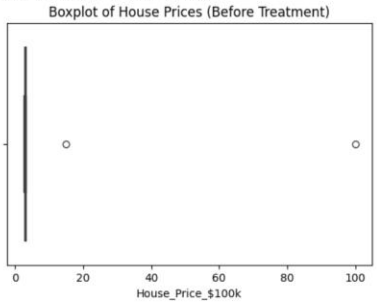


```
Corrected Data:

   House_Price_$100k   Z_score  Outlier_Z  Log_Price
0               2.5 -0.413529      False   0.916291
1               3.0 -0.397018      False   1.098612
2               2.8 -0.403622      False   1.029619
3               3.2 -0.390413      False   1.163151
4              15.0 -0.000734      False   2.708050
```

Variables Terminal 14:01 Python 3

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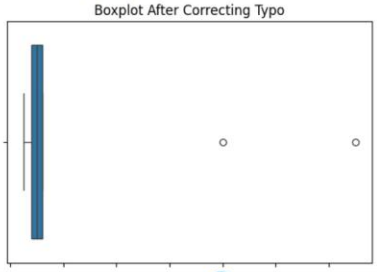
```
Corrected Data:

   House_Price_$100k   Z_score  Outlier_Z  Log_Price
0               2.5 -0.413529      False   0.916291
1               3.0 -0.397018      False   1.098612
2               2.8 -0.403622      False   1.029619
3               3.2 -0.390413      False   1.163151
4              15.0 -0.000734      False   2.708050
5               2.9 -0.400320      False   1.064711
6               3.1 -0.393715      False   1.131402
7               2.7 -0.406925      False   0.993252
8              10.0  2.806275      False   2.302585
```

Boxplot After Correcting Typo



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